Exercise 1: Population density and the MAUP; spatial quality of building data in the Helsinki region

Aderemi Adekanmi | 103573573 | October 2026

1.0 Problem 1 - Task 1.1

Data: Paavo postal code areas 2026 and the 1 km x 1 km population grid 2025 (Statistics Finland) and municipal borders 2025 (National Land Survey), all from Paituli. Helsinki, Espoo, Vantaa and Kauniainen were saved as the Capital Region layer, and Select By Location (Intersect) selected 185 postal code areas and 805 grid cells. For the postal areas, density per km² was calculated in the symbology as he_vakiy / (pinta_ala / 1,000,000). The grid cells are already 1 km² (vaesto). Both layers use five classes.

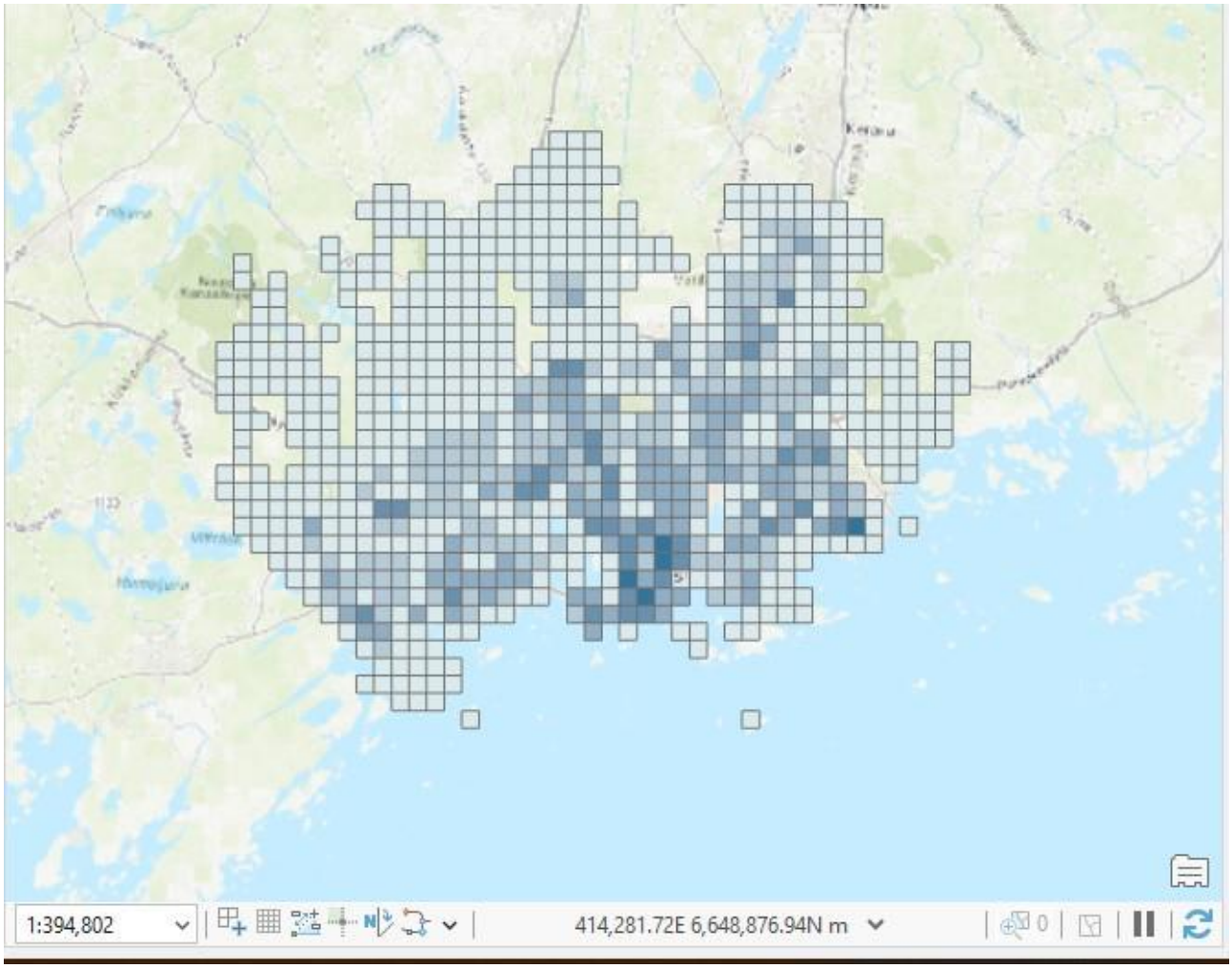


Figure 1. Population density (persons per km²) by postal code area, Helsinki Capital Region (Natural Breaks, five classes).

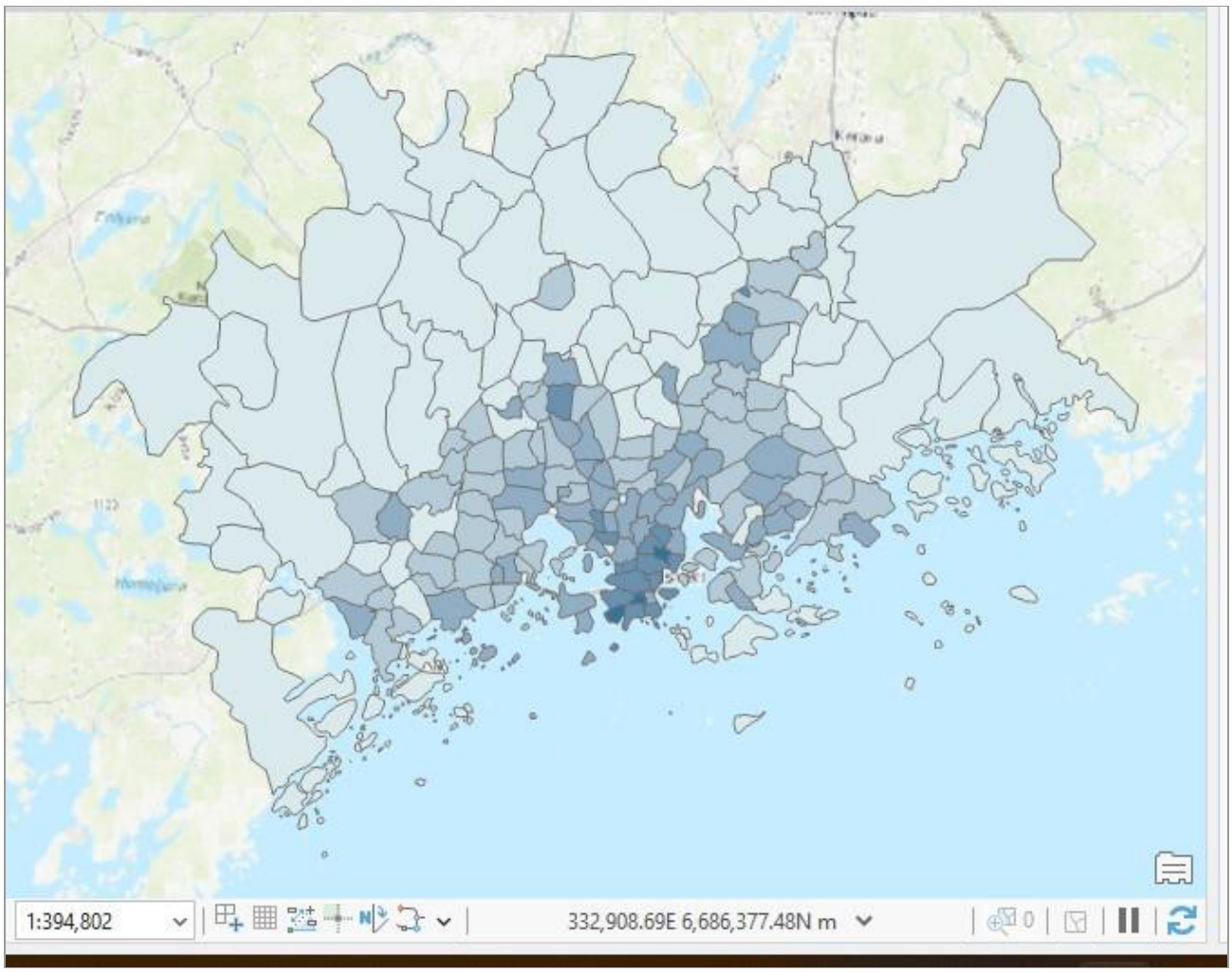


Figure 2. Population density (persons per km²) in 1 km x 1 km grid cells, with the same classes, limits and colours as Figure 1.

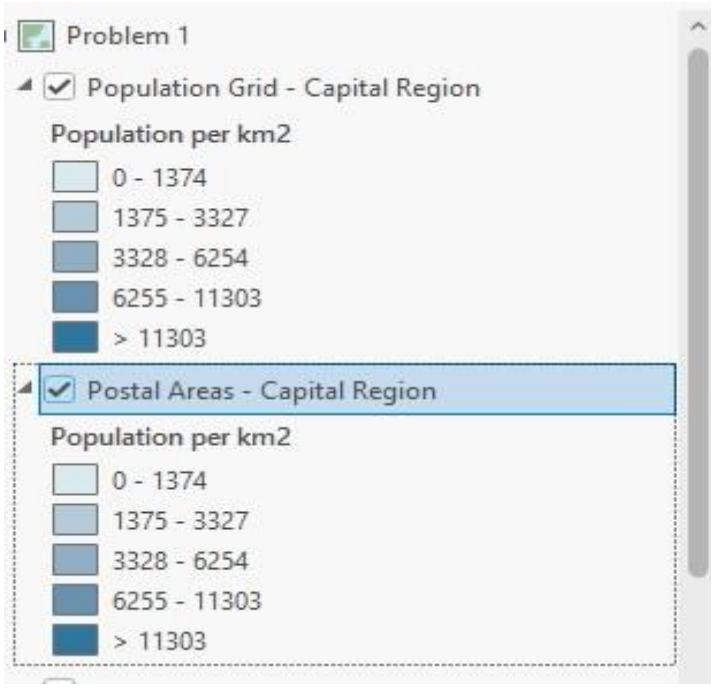


Figure 3. Legends of the grid map (top) and the postal code area map (bottom): identical classes, labels and colours.

Problem 1 - Task 1.2

Q1.1 How the two maps were made comparable

Both maps show persons per km² with the same classification and symbology. For the postal areas, density was calculated as population divided by area in km², and Natural Breaks (Jenks) with five classes gave the upper limits 1,374, 3,327, 6,254, 11,303 and 19,965. Natural Breaks run separately on the grid would give different limits, so I entered the same limits manually for the grid (its highest value, about 19,562, is below the top limit). Both layers also use the same colour ramp, labels and legend title, and the maps are shown at the same extent and scale (1:394,802).

Q1.2 Most important similarities and differences

Both maps show the same overall pattern: the highest densities are in the inner part of Helsinki on the coast, with a few higher-density clusters further north along the urban corridors, and the lowest class dominates the west, north and east. The main difference is the unit. Postal areas are small in the centre and very large in the periphery, where they appear as uniform blocks, while the grid shows variation within the same areas. The postal map also covers the whole territory, including sparsely populated land, whereas the grid has cells only where people live, so it has gaps and covers a smaller area. Postal boundaries are irregular, and the grid squares have equal area, which makes density easier to compare between places.

Q1.3 Is this a good example of the Modifiable Areal Unit Problem?

Yes. The same population data aggregated into two zoning systems gives different patterns, even though I used the same variable and classification. This shows both the scale effect (postal areas are much larger than grid cells in the periphery and smooth out variation) and the zoning effect (different boundaries put the same people into different units) described by Openshaw (1984) and Fotheringham & Wong (1991). Conclusions about where density is high therefore depend on the units chosen.

Problem 2 - Task 2.1

Data and method. HKI (layer Rakennukset_alue) and HSY (pks_rakennukset_paivittuva) were fetched through WFS with the feature limit raised to 200,000. The OSM buildings come from the Geofabrik Finland shapefile extract finland-260929-free.shp.zip (data as of 29 September 2026).

1. OSM and HSY buildings intersecting the Helsinki boundary were exported as local subsets (buildings crossing the border are included, which may slightly inflate the extras near the border), and HKI was copied locally. OSM was reprojected from WGS 84 to EUREF-FIN GK25FIN, the HKI system (transformation EUREF-FIN_to_WGS_84_1).
2. Repair Geometry (OGC) was run on all three layers; the feature counts were unchanged.
3. Erase was run in both directions (HKI minus the other data set, and the other data set minus HKI).

4. Erase outputs contain thin slivers where outlines differ slightly (69,165 of the 69,765 OSM buildings left a remnant), so missing and extra buildings were counted with Select By Location (inverted Intersect, i.e. no overlap in the other data set), and areas with Summary Statistics.

Data set	Local copy used	Buildings	Total area (m²)
HKI	HKI buildings (HKI_Local, reference)	84,939	19,083,721
HSY	HSY_Helsinki	48,889	16,837,411
OSM	OSM_Helsinki_Project (reprojected)	69,765	21,496,633

Table 1. Data sets used in Problem 2 after processing.

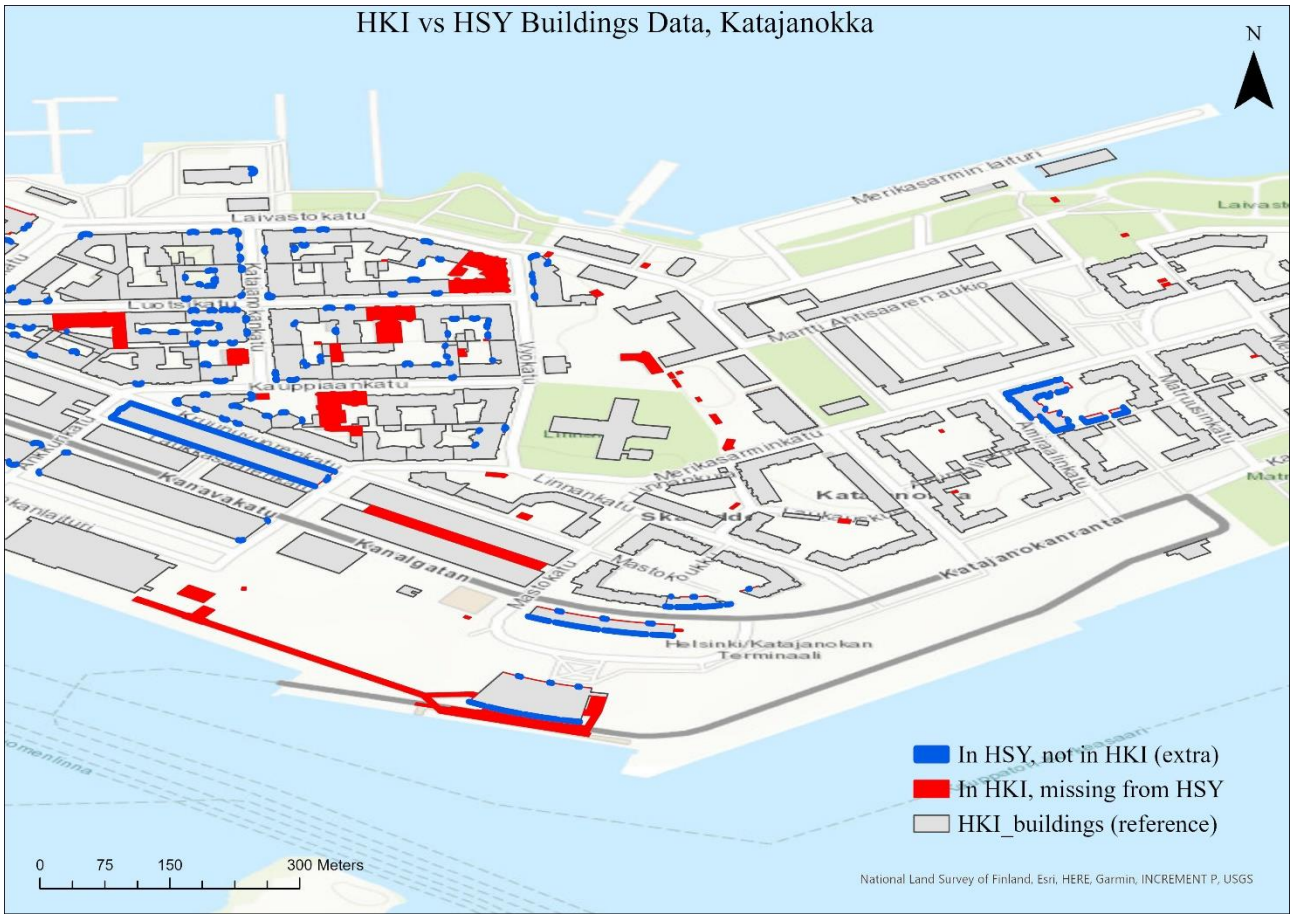


Figure 4. Differences between HKI and HSY building data, Katajanokka, Helsinki (about 1:5,000). Grey: HKI buildings (reference). Red: parts of HKI buildings with no HSY building (missing from HSY). Blue: parts of HSY buildings with no HKI building (extra in HSY). Derived with Erase after geometry repair; the blue dots and thin lines along building edges are slivers from small outline differences.

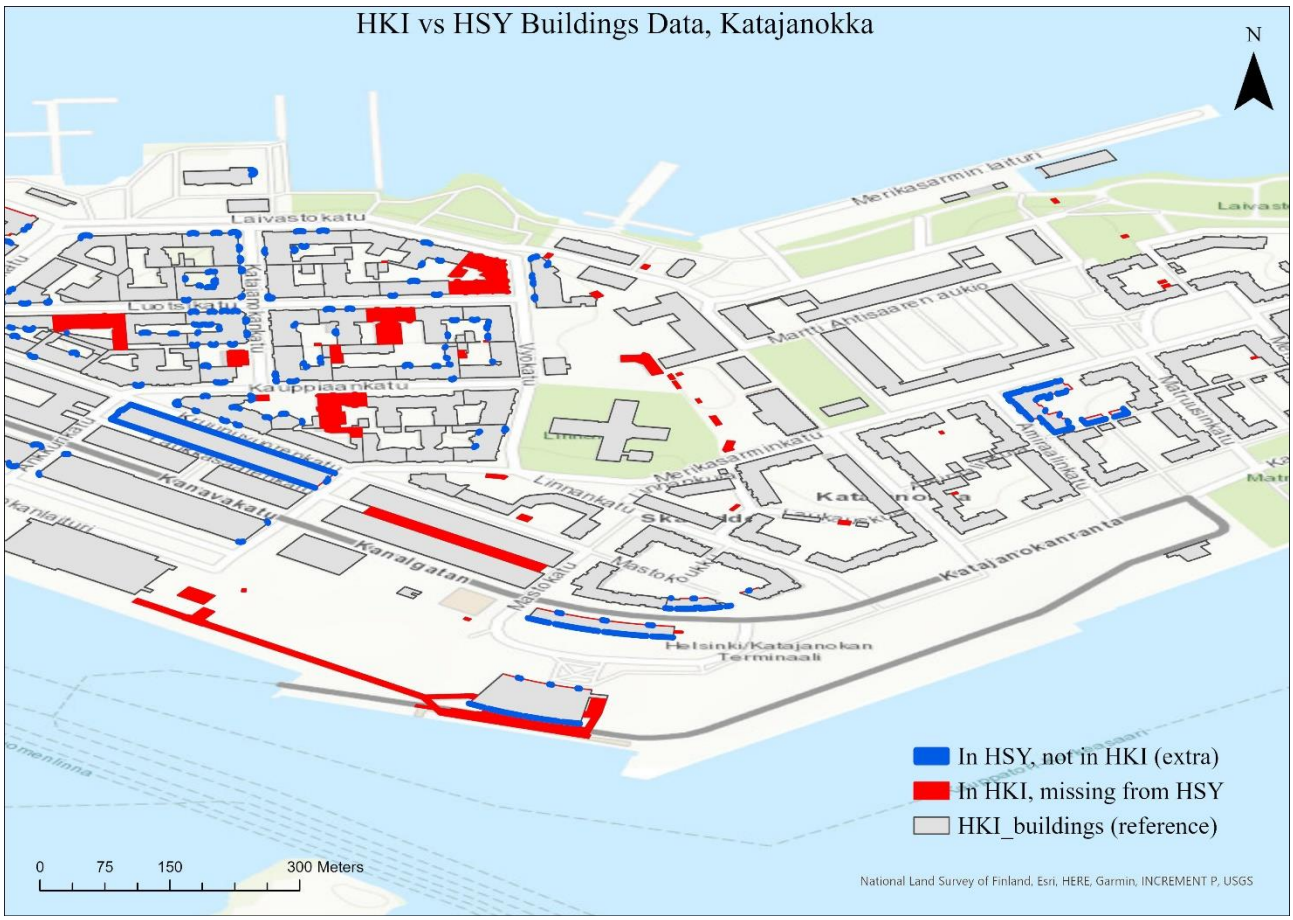


Figure 5. Differences between HKI and OSM building data, Katajanokka, Helsinki (about 1:5,000). Grey: HKI buildings (reference). Red: parts of HKI buildings with no OSM building (missing from OSM). Blue: parts of OSM buildings with no HKI building (extra in OSM). Derived with Erase after geometry repair and reprojection; the thin red and blue lines along most edges are slivers from small offsets between the outlines. A few solid blue patches are OSM shapes with no HKI counterpart.

Problem 2 - Task 2.2

Measure	HKI (reference)	HSY	OSM
Buildings	84,939	48,889	69,765
Total area (m²)	19,083,721	16,837,411	21,496,633
HKI buildings missing (count)	n/a	33,194 (39.1%)	17,152 (20.2%)
Extra buildings, not in HKI (count)	n/a	277 (0.6%)	6,589 (9.4%)
HKI area missing (m²)	n/a	2,544,634 (13.3%)	972,680 (5.1%)
Extra area, not in HKI (m²)	n/a	287,946 (1.7%)	3,269,947 (15.2%)

Table 2. Completeness of HSY and OSM compared with HKI (shares of HKI for missing, of the data set itself for extra).

Q2.1 Spatial accuracy and completeness of HSY

HSY is incomplete, but what it contains is reliable. 39.1% of the HKI buildings (33,194) have no HSY counterpart, but they are mostly small, so only 13.3% of the HKI building area is missing, and some large buildings are also absent (Figure 4). Only 277 HSY buildings (0.6%) have no HKI counterpart. The positional accuracy appears good: in Katajanokka the HSY outlines follow HKI closely, the differences are thin slivers, and the overlap is about 85% of the combined area. The biggest problem is therefore omission. A possible

reason is that HSY lists only completed buildings with their own entrance that have had a final inspection (HSY, n.d.), and 39.8% of the HKI features (33,816) are outbuildings, a share close to the omission rate, but I did not test this overlap directly.

Q2.2 Spatial accuracy and completeness of OSM

OSM is more complete than HSY, but it has more extra data and less consistent outlines. 20.2% of the HKI buildings (17,152) are missing, but only 5.1% of the HKI area, so OSM covers about 95% of it. 6,589 OSM buildings (9.4%) have no HKI counterpart and 15.2% of the OSM area is extra, so the OSM total area (21.5 million m²) exceeds that of HKI (19.1 million m²). The area figure is an upper bound because it includes edge slivers, and some extras are probably structures that HKI does not hold, as HKI excludes buildings under 10 m² and negligible structures without a building identifier (City of Helsinki, n.d.). In Katajanokka the OSM outlines line up with HKI without a systematic shift, but thin lines along most edges show small offsets (overlap about 81%). The main problems are the extras and the inconsistent outlines, which is typical of volunteered data (Haklay, 2010; Goodchild & Li, 2012), and the remaining omissions.

Q2.3 Which data set is of higher quality, OSM or HSY?

I consider OSM to be of higher quality, using the ISO 19157 quality elements of completeness (omission and commission) and positional accuracy (ISO, 2013). By building count OSM finds 79.8% of the HKI buildings against 60.9% for HSY, and the harmonic mean of completeness and correctness (F-score) is 0.85 against 0.76. Omission is the more serious error for most uses, such as estimating the building stock or energy demand, because missing buildings cannot be recovered while extra ones can be filtered out, and the OSM commission error is probably overestimated (slivers and the HKI specification). Positional agreement is good for both. The case is not clear-cut, however: by area HSY scores slightly higher (F-score 0.92 against 0.90) and has almost no extras, so HSY would be preferable where false buildings cannot be tolerated.

Q2.4 Was it reasonable to use HKI as the baseline?

It was a reasonable assumption, but not a perfect one. HKI is based on the base map of the City Survey Services, mapped at a scale of 1:500, and the city has the mandate and the need to keep it up to date (City of Helsinki, n.d.), so it is the best available reference here. It is, however, a reference and not the truth (ISO, 2013). It follows its own specification and does not contain buildings under 10 m² or negligible structures such as canopies and sheds without a building identifier, so some extras in HSY and OSM are errors only relative to HKI. It also contains polygons that are not ordinary buildings: the long waterfront strip in Katajanokka that is missing from both HSY and OSM is a valid record of type “other use” (Muu käyttötarkoitus) without a RATU identifier or address, and 9.3% of the HKI features (7,869) have this type, so the omission rates are probably slightly overstated. Differences in dates, and the overlap method, which cannot tell whether overlapping polygons are the same real-world building (a new building on the plot of a demolished one counts as a match), add uncertainty, and HKI covers only Helsinki. A check against aerial imagery would give a stronger validation.

Problem 3

Time spent: about 10 hours. My background is in Real Estate Economics, and I was introduced to ArcGIS in the course From Measurement to Maps, so I am still developing my skills in the software, and the time would probably have been shorter for someone more efficient with the tools. Most of the time went to understanding the data and how the tools behave.

Feedback: the exercise made the MAUP and the data quality concepts concrete. I have three suggestions for the instructors. (1) The instructions could mention that WFS layers are read-only web connections: Select By Location on the HSY layer failed for me with ERROR 160228 until I exported a local copy. (2) They could warn that Erase creates many slivers, so the number of output features is not the number of missing buildings. (3) They could offer a fallback for the Geofabrik download, which gave me proxy errors and a 404 on the "latest" link**.**

Acknowledgement

I used an AI assistant for step-by-step guidance in ArcGIS Pro. I ran the analysis, produced the maps and checked the results myself.

References

City of Helsinki (n.d.). Buildings in Helsinki (Helsingin rakennukset). Helsinki Region Infoshare. https://hri.fi/data/en_GB/dataset/helsingin-rakennukset

Fotheringham, A. S., & Wong, D. W. S. (1991). The modifiable areal unit problem in multivariate statistical analysis. Environment and Planning A, 23(7), 1025–1044.

Goodchild, M. F., & Li, L. (2012). Assuring the quality of volunteered geographic information. Spatial Statistics, 1, 110–120.

Haklay, M. (2010). How good is volunteered geographical information? A comparative study of OpenStreetMap and Ordnance Survey datasets. Environment and Planning B, 37(4), 682–703.

HSY (n.d.). Buildings in the Helsinki metropolitan area. Helsinki Region Infoshare. https://hri.fi/data/en/dataset/paakaupunkiseudun-rakennukset

ISO (2013). ISO 19157:2013 Geographic information – Data quality. International Organization for Standardization.

Openshaw, S. (1984). The Modifiable Areal Unit Problem. CATMOG 38. Geo Books, Norwich.

Data: Statistics Finland and National Land Survey of Finland via Paituli (CSC); City of Helsinki and HSY WFS interfaces; OpenStreetMap contributors (Geofabrik extract, ODbL).